

Ahan Bhatt

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SUMMARY

Recent M.S. graduate in Computer Science and Engineering with experience building transformer compilers, GPU-accelerated LLM inference systems, adversarial evaluation infrastructure, and trustworthy AI platforms. Strong background in Python, C++, CUDA, Triton, PyTorch, FastAPI, and full-stack ML systems, supported by research experience and publications across LLM evaluation, explainable AI, time-series modeling, and applied machine learning.

EDUCATION

University of California, Santa Cruz [3.82 GPA]
Master of Science in Computer Science and Engineering

Santa Cruz, California
Aug. 2025 – Jun 2026

Indus Institute of Technology and Engineering [3.94 GPA]
Bachelor of Technology in Computer Engineering

Ahmedabad, India
Aug. 2020 – July 2024

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R

Machine Learning: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, NumPy, Pandas, Matplotlib

ML Systems and Compilers: CUDA, Triton, C++20, pybind11, custom IRs, operator fusion, High-Performance Computing (HPC), kernel selection, memory/execution planning, paged KV caching, quantization

LLM and Data Tools: LangChain, LangGraph, MCP, RAG, Hybrid RAG, ChromaDB, vector databases, Multi-Agent Systems, Agentic AI, Tool Calling, LoRA/QLoRA, LLM Evaluation, LLM Observability, AI Guardrails, Human-in-the-Loop (HITL), PyPDF

Backend and Deployment: Node.js, FastAPI, Flask, REST APIs, FHIR, Docker, CMake, Bazel, Kubernetes, Linux, Git, CI/CD, Nginx

Cloud: AWS EC2, S3, IAM, Lambda, SageMaker, CloudWatch

Data Engineering: PostgreSQL, Redis, Real-Time Data Pipelines, ETL, schema design

Software Engineering: TDD, API integration, DSA, system design

EXPERIENCE

Researcher
[*AI Explainability and Accountability \(AIEA\) Lab @ UCSC*](#)

September 2025 – June 2026
Santa Cruz, California

- Built ReliaGuard Studio and BreakPoint with Next.js, TypeScript, FastAPI, PostgreSQL, Docker, and MLflow to detect over/underreliance in human-AI decisions and compile real LLM failure modes into adversarial eval suites using neuro-symbolic modeling, conformal gating, multi-judge validation, and explainable dashboards.

Data Scientist
[*Growerxx*](#)

August 2024 – August 2025
Ahmedabad, Gujarat, India

- Built a healthcare prior-authorization and clinical workflow agent that used FHIR, hybrid RAG, LangGraph, tool integrations (including MCP), automated evaluations, and clinician approval gates to retrieve and validate patient evidence while supporting HIPAA-compliant workflows. Built production multi-agent systems for a retail/e-commerce client that integrated CRM, inventory, and customer-support platforms to automate service, product recommendations, and order resolution, with observability, guardrails, and human escalation.

AI Researcher (Part-time)
[*AI Institute of South Carolina*](#)

June 2024 – May 2025
Columbia, South Carolina (remote)

- Conducted research on multimodal toxicity mitigation and anomaly detection, using PostgreSQL-backed experiment tracking to manage dataset metadata, model outputs, evaluation metrics, and benchmarking results for projects like DE-HATE and Time Series Foundation Model evaluation.

Data Scientist
[*Space Applications Centre, ISRO*](#)

January 2024 – May. 2024
Ahmedabad, Gujarat, India

- Built an end-to-end LSTM satellite clock-bias prediction platform with Apache Kafka live streaming and a Next.js/React/TypeScript UI, automating preprocessing and anomaly filtering to improve navigation timing forecasts.

AI And Data Engineer
[*Intellytics*](#)

February 2022 – January 2024
Ahmedabad, Gujarat, India

- Coordinated with multiple US-based clients to build agentic AI, LLM, and ML solutions using RAG, tool calling, hybrid retrieval, LoRA/QLoRA, Kubernetes, real-time data pipelines, and production evaluation/observability.

PROJECTS

CacheIR [CUDA, Triton, C++20, custom compiler IRs, PyTorch, CMake, pybind11, FastAPI, NumPy]

- A narrow, inspectable compiler/runtime for GPU-accelerated decoder-only transformer inference, up to 3.24× faster than vLLM on certain workloads. It imports Llama/Mistral/Qwen models into a custom IR, applies transformer-specific compiler optimizations, plans memory/execution, selects CUDA kernels, and serves through a reference runtime.

BreakPoint [LLM evaluation pipelines, multi-judge validation, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, MLflow, DuckDB]

- A failure-to-eval compiler for LLM systems that mines real failure modes into versioned adversarial regression suites with DSL-defined traps/rubrics, multi-judge calibration, RAG/tool simulators, CI gates, FastAPI/Next.js dashboards, and 360 validated eval cases across 9 failure families.

ReliaGuard Studio [Neuro-symbolic AI, conformal prediction, FastAPI, Next.js, TypeScript, PostgreSQL, MLflow, Docker, scikit-learn]

Research Forge – A Self-Improving Research Agent [LangGraph, autonomous agents, Neo4j, OpenAI API, Pydantic, Streamlit, Python]

Meeting AI Assistant [Real-time multimodal AI, Electron, React, OpenAI API, Node.js, Express, TypeScript]

Time Series Foundation Model Evaluation for Anomaly Detection [TimeGPT, Chronos, Time-MOE, PyTorch, TensorFlow, scikit-learn, Pandas, Python]

Knowledge Graph Generation from Large Language Models [GraphRAG, GPT-4, LLaMA-2, BERT, Transformers, Python]

Leveraging LSTM for Predictive Modeling of Satellite Clock Bias

- *Presented at [CDMA 2025 conference](#) and available on [IEEE Xplore](#)*

Generating Knowledge Graphs from Large Language Models

Time Series Foundational Models: Their Role in Anomaly Detection and Prediction

- *Presented at [AAAI 2025 Anomaly Detection in Scientific Domains Workshop](#)*

Med-Bot: An AI-Powered Assistant to Provide Accurate and Reliable Medical Information